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NARULA INSTITUTE OF TECHNOLOGY
AN AUTONOMOUS INSTITUTE UNDER MAKAUT
2025
2024-25 EVEN SEMESTER
CS201 - PROGRAMMING FOR PROBLEM SOLVING

TIME ALLOTTED: 3Hours**Maximum Marks : 70****Instructions to the candidate:***Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The C code 'for(;;)' represents an infinite loop. It can be terminated by _____ (1) C01 BL1 PO1
(a) break (b) exit(0)
(c) abort() (d) terminate
2. Which of the following are C preprocessors? (1) C01 BL1 PO1
(a) #ifdef (b) #define
(c) #endif (d) d) all of the mentioned
3. Which of the following c statement will calculate the correct size of an array of 10 integers? (1) C01 BL1 PO1
(a) sizeof(a[10]); (b) sizeof(*a);
(c) sizeof(a); (d) sizeof(&a);
4. What will be the output of the following C code? (1) C01 BL1 PO1

```
#include <stdio.h>
void main()
{
    double k = 0;
    for (k = 0.0; k < 3.0; k++)
        printf("Hello");
}
```


(a) Run time error (b) Hello is printed thrice

- (c) Hello is printed twice (d) Hello is printed infinitely

5. What will be the output of the following C code? (1) CO1 BL1 PO1

```
#include<stdio.h>
enum colour
{
    blue, red, yellow
};
main()
{
    enum colour c;
    c=yellow;
    printf("%d",c);
}
```

- (a) 1 (b) 2
(c) 0 (d) Error

6. What will be the output of the following C code? (1) CO1 BL1 PO1

```
#include <stdio.h>
int main()
{
    const int ary[4] = {1, 2, 3, 4};
    int *p;
    p = ary + 3;
    *p = 5;
    printf("%d\n", ary[3]);
}
```

- (a) 4 (b) 5
(c) Compile time error (d) 3

7. Which function reallocates memory? (1) CO1 BL1 PO1

- (a) realloc (b) calloc

(c) malloc (d) None of these

8. What is the purpose of a return statement in a C function? (1) CO1 BL1 PO1

- (a) To exit the program (b) To send a value back to the function caller
(c) To print a value (d) To terminate a loop

9. What is the output of this recursive function call fact(3)? (1) CO3 BL2 PO3
int fact(int n) { if(n==1) return 1; else return n*fact(n-1); }

- (a) 3 (b) 6
(c) 9 (d) 1

10. What is the output of: `int fun(int x){ if(x==0) return 0; return x+fun(x-1); } printf("%d", fun(3));` (1) CO1 BL1 PO1

- (a) 6 (b) 5
(c) 3 (d) 1

11. How is character 'A' represented in ASCII? (1) CO1 BL1 PO1

- (a) 41 (b) 61
(c) 65 (d) 97

12. What is the output of `int a[3] = {1,2,3}; printf("%d", *(a+1));`? (1) CO1 BL1 PO1

- (a) 1 (b) 2
(c) 3 (d) Error

Part B

(Answer any three of the following) 3x5=15

13. Answer the following :

- (a) Define algorithm. (2) CO1 BL1 PO1
(b) Write an algorithm and draw a flowchart to find the largest of three numbers. (3) CO2 BL3 PO1

14. Differentiate between primary and secondary memory with suitable examples. (5) CO1 BL2 PO1

15. Write a C program to find GCD of two numbers using ternary operator and for loop (5) CO3 BL1 PO1

16. Write a C program to demonstrate the use of arithmetic, relational, and logical operators. (5) CO2 BL3 PO3

17. Compare compiler, interpreter, and assembler with suitable examples. (5) CO2 BL1 PO1

Part C

(Answer any three of the following) 3x15=45

18. Answer the following :

- (a) Compare recursive and iterative approaches using a suitable example (4) CO1 BL4 PO1
(b) Define function prototype. (2) CO1 BL1 PO1

- | | | | | |
|--|-----|-----|-----|-----|
| (c) Discuss the function call, function definition and function prototype with examples | (5) | C01 | BL6 | PO1 |
| (d) Define the following with examples:
i) auto storage class
ii) static storage class | (4) | C02 | BL5 | PO1 |
19. (a) What are command-line arguments? (3) C05 BL1 PO1
- (b) Write the syntax and explain the purpose of the following file handling functions:

fopen()
fclose()
fgetc()
fputc() (4) C01 BL1 PO1
- (c) Write a C program to read characters from a file using fgetc() and display them on the screen. (5) C04 BL1 PO1
- (d) What is the difference between a structure and a union? Give at least three differences with examples. (3) C03 BL1 PO1
20. Answer the following :
- | | | | | |
|--|-----|-----|-----|-----|
| (a) Explain how pointers and arrays are related with example | (5) | C05 | BL2 | PO1 |
| (b) What is preprocessor directive? | (3) | C01 | BL1 | PO1 |
| (c) Explain #define and #include preprocessor directives | (4) | C01 | BL3 | PO1 |
| (d) State how the pointer variable declared and initialized? | (3) | C03 | BL2 | PO1 |
21. (a) Define a structure. How is a structure variable declared and initialized? (5) C04 BL4 PO2
- (b) Write a C program to read and display information of 3 students using an array of structures. (5) C04 BL3 PO4
- (c) Explain how structure and pointer can be used together. Give example. (5) C04 BL2 PO2
22. Answer the following :
- | | | | | |
|---|-----|-----|-----|-----|
| (a) Design a C program to find the reverse of an integer number and check whether it is palindrome or not | (8) | C01 | BL6 | PO1 |
| (b) Compare between formal parameters and actual parameters with an example | (3) | C03 | BL5 | PO1 |

- (c) Write a C Program to implement string copy operation
STRCOPY(str1,str2)

(4) C02 BL1 PO1